# OP SILENT PERIMETER

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# PHASE 1 — Routing Crisis (Cisco Packet Tracer)

**1.1 Network Topology & IP Addressing.** The following topology in Cisco Packet Tracer with three routers (R1, R2, R3) and their respective LANs was designed:

| Router | LAN Network | Transit Link | Protocol |
|---|---|---|---|
| R1 (HQ) | 10.10.10.0/24 | R1↔R2: 192.168.12.0/30<br>R1↔R3: 192.168.13.0/30 | OSPF Area 0 |
| R2 (Operations) | 20.20.20.0/24 | R2↔R3: 192.168.23.0/30 | OSPF Area 0 |
| R3 (Logistics) | 30.30.30.0/24 | Connected to R1 & R2 | EIGRP AS 100 |

## 1.2 Interface IP Configuration Commands

### Router R1:

```
R1(config)# interface fastethernet0/0
R1(config-if)# ip address 10.10.10.1 255.255.255.0
R1(config-if)# no shutdown
R1(config)# interface Serial0/0/0   ! Link to R2
R1(config-if)# ip address 192.168.12.1 255.255.255.252
R1(config-if)# no shutdown
R1(config)# interface Serial0/0/1   ! Link to R3
R1(config-if)# ip address 192.168.13.1 255.255.255.252
R1(config-if)# no shutdown
```

### Router R2:

```
R2(config)# interface fastEthernet0/0
R2(config-if)# ip address 20.20.20.1 255.255.255.0
R2(config-if)# no shutdown
R2(config)# interface Serial0/0/0   ! Link to R1
```

```
R2(config-if)# ip address 192.168.12.2 255.255.255.252
R2(config-if)# no shutdown
R2(config)# interface Serial0/0/1   ! Link to R3
R2(config-if)# ip address 192.168.23.1 255.255.255.252
R2(config-if)# no shutdown
```

### Router R3:

```
R3(config)# interface fastEthernet0/0
R3(config-if)# ip address 30.30.30.1 255.255.255.0
R3(config-if)# no shutdown
R3(config)# interface Serial0/0/0   ! Link to R1
R3(config-if)# ip address 192.168.13.2 255.255.255.252
R3(config-if)# no shutdown
R3(config)# interface Serial0/0/1   ! Link to R2
R3(config-if)# ip address 192.168.23.2 255.255.255.252
R3(config-if)# no shutdown
```

## 1.3 OSPF Configuration (R1 and R2)

### R1 OSPF:

```
R1(config)# router ospf 1
R1(config-router)# network 10.10.10.0 0.0.0.255 area 0
R1(config-router)# network 192.168.12.0 0.0.0.3 area 0
R1(config-router)# network 192.168.13.0 0.0.0.3 area 0
```

### R2 OSPF:

```
R2(config)# router ospf 1
R2(config-router)# network 20.20.20.0 0.0.0.255 area 0
R2(config-router)# network 192.168.12.0 0.0.0.3 area 0
R2(config-router)# network 192.168.23.0 0.0.0.3 area 0
```

## 1.4 EIGRP Configuration (R3)

R3(config)# router eigrp 100

R3(config-router)# network 30.30.30.0 0.0.0.255

R3(config-router)# network 192.168.13.0 0.0.0.3

R3(config-router)# network 192.168.23.0 0.0.0.3

R3(config-router)# no auto-summary

## 1.5 Route Redistribution

### On R1 — Redistribute EIGRP into OSPF:

R1(config)# router ospf 1

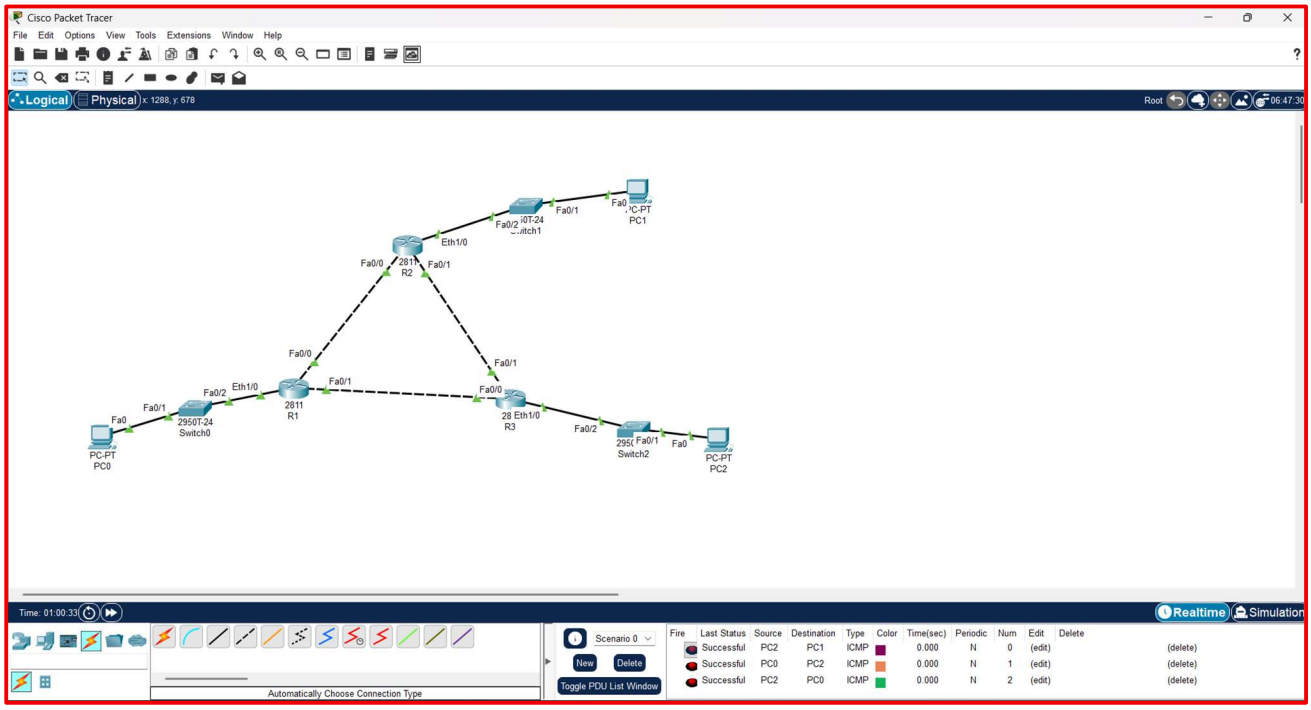
R1(config-router)# redistribute eigrp 100 subnets

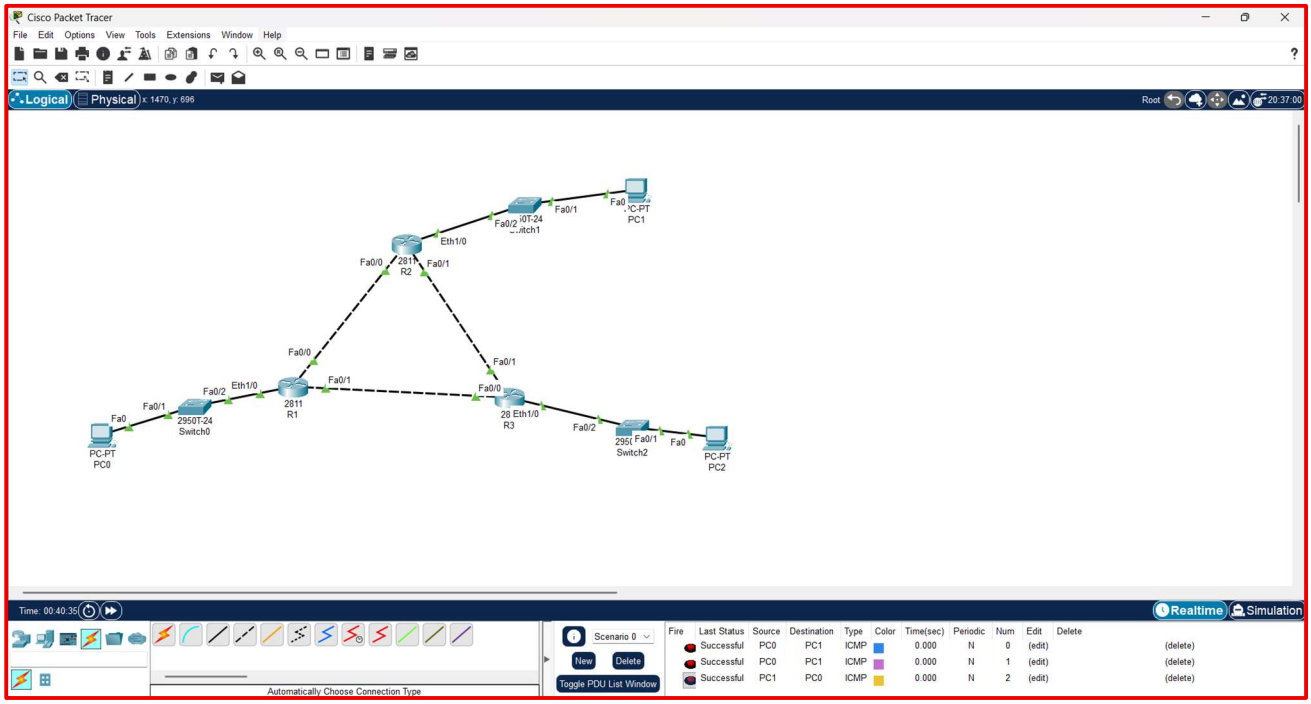
### On R1 — Redistribute OSPF into EIGRP:

R1(config)# router eigrp 100

R1(config-router)# network 192.168.13.0 0.0.0.3

R1(config-router)# redistribute ospf 1 metric 10000 100 255 1 1500

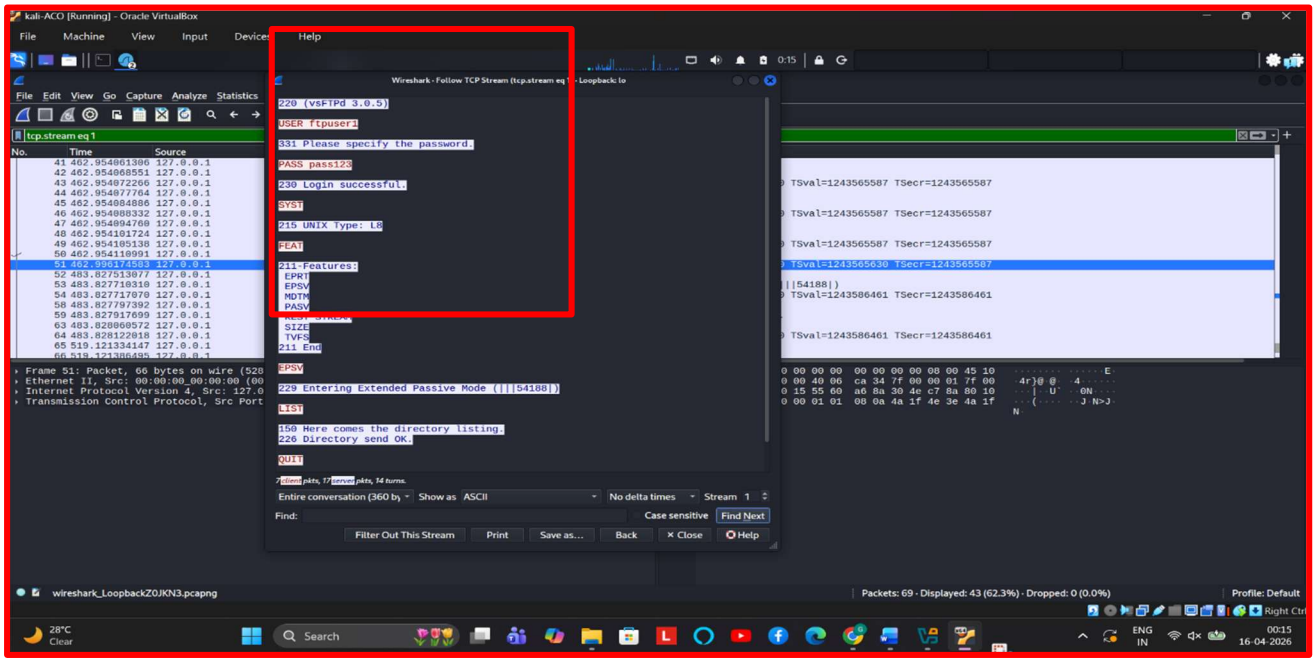



Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1470, y: 696
Root
20:37:00
PC-PT
PC0
2950T-24
Switch0
2811
R1
2811
R2
28 Eth1/0
R3
2950T-24
Switch1
2950T-24
Switch2
PC-PT
PC1
PC-PT
PC2
Fa0
Fa0/1
Fa0/2
Eth1/0
Fa0/0
Time: 00:40:35
Realtime
Simulation
Automatically Choose Connection Type
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC0 PC1 ICMP 0.000 N 0 (edit) (delete)
Successful PC0 PC1 ICMP 0.000 N 1 (edit) (delete)
Successful PC1 PC0 ICMP 0.000 N 2 (edit) (delete)

# PHASE 2 — Intercepted Credentials (Wireshark)

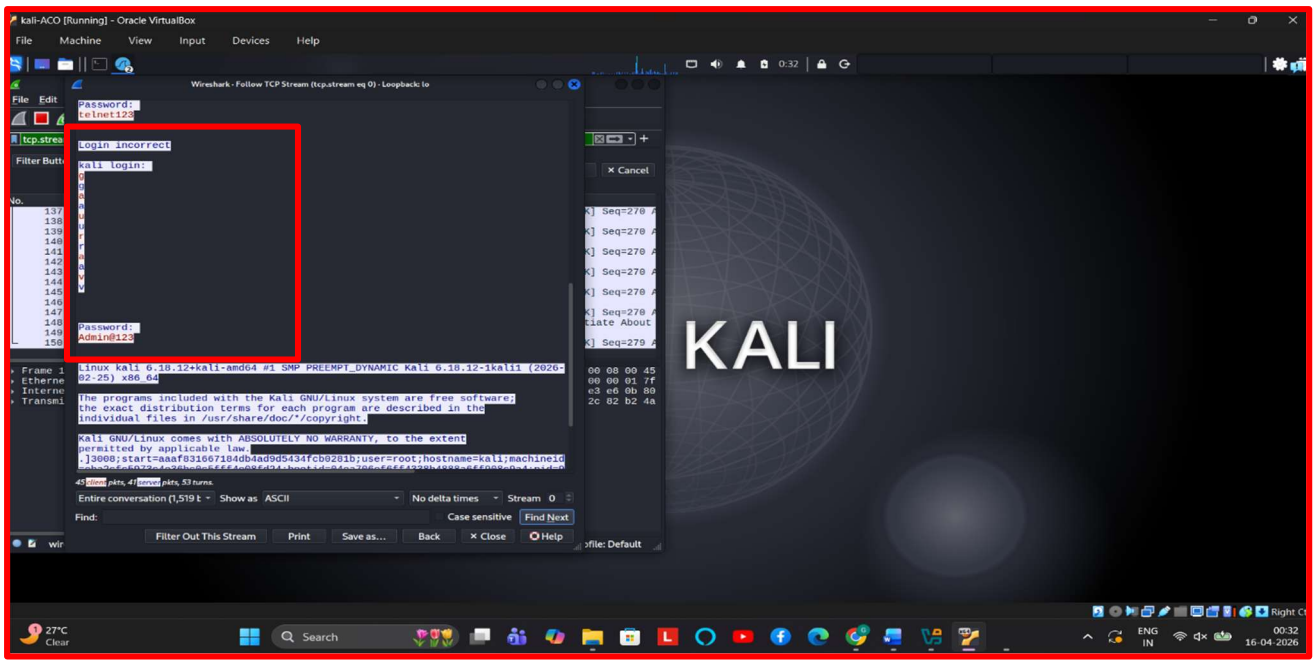
## 2.1 FTP Credential Capture

| Field | FTP | Telnet | HTTP (DVWA) |
|---|---|---|---|
| Username | ftpuser1 | root | admin |
| Password | Pass123 | root | password |
| Source IP | 10.10.10.50 | 10.10.10.60 | 10.10.10.70 |
| Destination IP | 10.10.10.10 (Server) | 30.30.30.10 | 20.20.20.10 (DVWA) |




PH2- FTP- SUCCESSFUL CAPTURE OF FTPUSER AND PASSWORD

## 2.2 Telnet Credential Capture

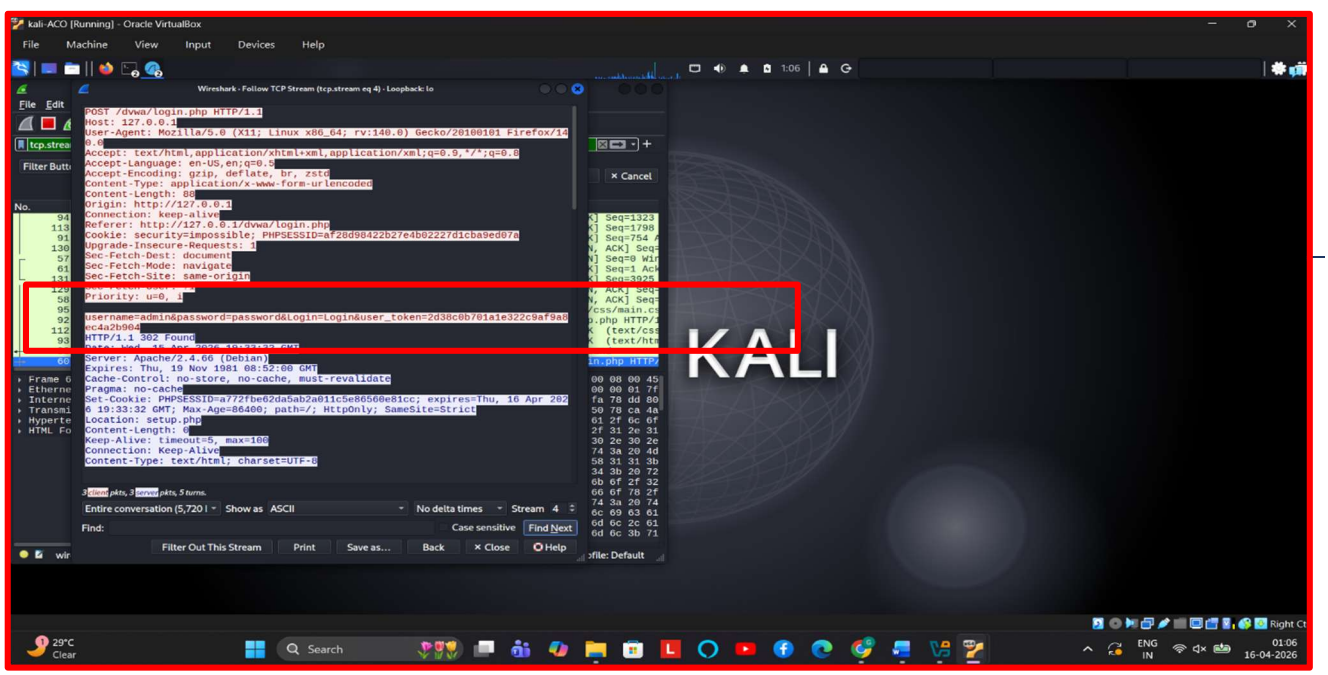



PH2- TELNET - CAPTURE OF TELNET USER AND PASSWORD

## 2.3 DVWA HTTP POST Capture

HTTP POST requests send form data unencrypted. Apply filter:

`http.request.method == POST`




PH2- DVWA - CAPTURE OF USERID AND PASSWORD

# PHASE 3 — Intrusion Detection (Suricata)

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## 3.1 Suricata Installation

```
sudo apt update && sudo apt install suricata -y
sudo suricata-update                # Update rule signatures
sudo systemctl enable suricata
```

## 3.2 Custom Rules File

**Edit: /etc/suricata/rules/local.rules**

### Rule 1 — ICMP Recon (Ping Detection):

```
alert icmp any any -> any any (msg:"RECON: ICMP Ping Detected"; itype:8; sid:1000001; rev:1;)
```

### Rule 2 — Nmap SYN Scan Detection:

```
alert tcp any any -> any any (msg:"SCAN: Nmap SYN Scan Detected"; flags:S,12; threshold: type both, track by_src, count 20, seconds 3; sid:1000002; rev:1;)
```

### Rule 3 — Telnet Usage Alert:

```
alert tcp any any -> any 23 (msg:"POLICY: Telnet Usage Detected - Insecure Protocol"; sid:1000003; rev:1;)
```

### Rule 4 — FTP Credential Exposure:

```
alert tcp any any -> any 21 (msg:"CREDS: FTP USER Command Detected"; content:"USER"; nocase; sid:1000004; rev:1;)
alert tcp any any -> any 21 (msg:"CREDS: FTP PASS Command Detected"; content:"PASS"; nocase; sid:1000005; rev:1;)
```

### Rule 5 — Port Scan Detection (Multiple Ports):

```
alert tcp any any -> any any (msg:"SCAN: Multiple Port Scan from Same Source"; flags:S; threshold: type threshold, track by_src, count 5, seconds 10; sid:1000006; rev:1;)
```

## 3.3 Enable Local Rules in suricata.yaml

```
sudo nano /etc/suricata/suricata.yaml
Locate the rule-files section and add:
  - /etc/suricata/rules/local.rules
```

## 3.4 Testing Commands

Test ICMP Rule:
```
ping -c 5 192.168.12.1
```

Test Nmap SYN Scan Rule:
```
nmap -sS 192.168.12.1
```

Test Telnet Rule:
```
telnet 30.30.30.10 23
```

Test FTP Rule:
ftp 10.10.10.10

**Run Suricata in test mode:**

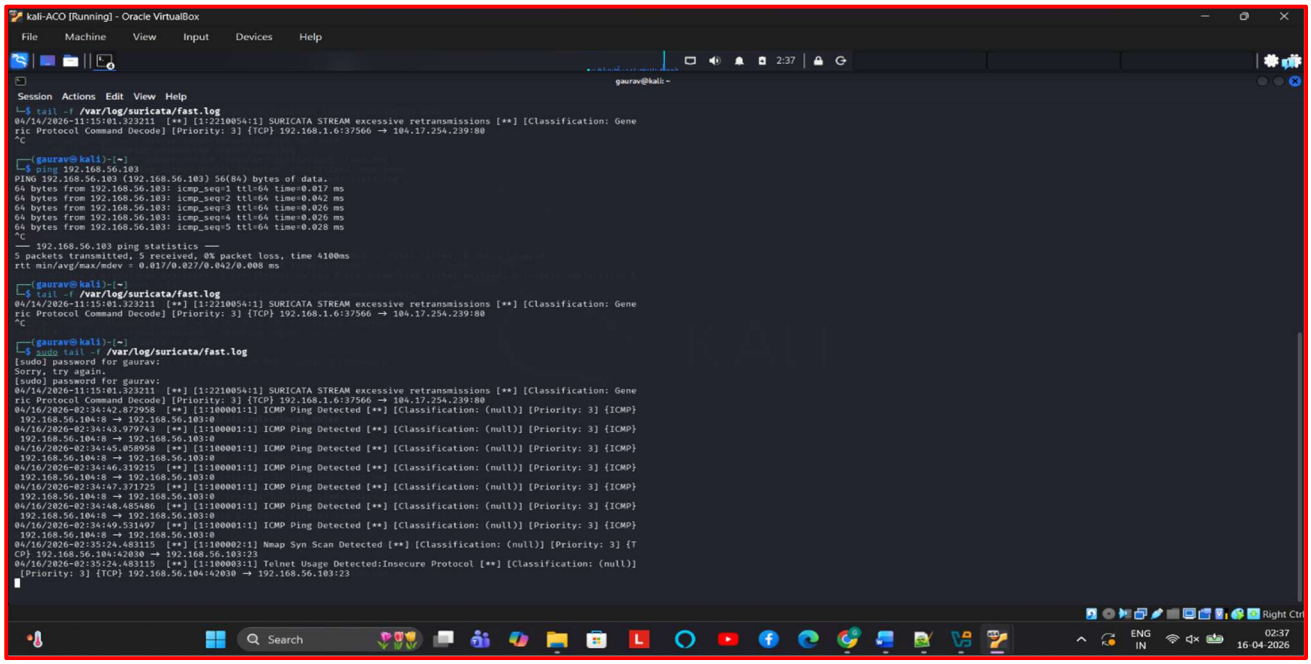
sudo suricata -c /etc/suricata/suricata.yaml -i eth0

**Watch alerts live:**

sudo tail -f /var/log/suricata/fast.log

## 3.5 Highest Priority Alert. Rule 5 (Port Scan / Multiple Port Hits)

3.5.1 Port scanning is reconnaissance — it precedes every intrusion.
3.5.2 It has the broadest implication: attacker is mapping the entire network.
3.5.3 Should trigger immediate defensive lockdown of the source IP.
3.5.4 ICMP pings alone may be benign; a methodical port scan is clearly hostile in a military network.



# PHASE 4 — Host Forensics (Netstat)

## 4.1 Commands and Their Purpose

| Command | Purpose | Key Info |
|---|---|---|
| netstat -an | All connections, no DNS resolution | Shows numeric IPs and ports |
| netstat -ano | All connections + PID | Links connection to process ID |
| netstat -rn | Routing table (numeric) | Default gateway, routes |
| netstat -tulpn | TCP/UDP listening ports + process name | Linux — reveals which services are open |

# PHASE 5 — Final War Scenario: Incident Report

## Incident Timeline (Reconstructed)

| Time | Event | Evidence |
|---|---|---|
| 2247 | Attacker begins Nmap SYN scan | Suricata Rule 5 — port scan alert in fast.log |
| 2255 | Routing to Logistics disrupted | Phase 1 — OSPF wildcard mask manipulated |
| 2258 | Telnet credentials stolen | Wireshark — plaintext USER/PASS in capture |
| 2301 | FTP data transfer observed | Wireshark — RETR/STOR commands logged |
| 2305 | Laptop connects to port 4444 | Netstat — ESTABLISHED to 203.0.113.99:4444 |
| 2310 | C2 beacon confirmed active | Process PID 3421 — Meterpreter payload |

**1. What Happened First.** The Nmap reconnaissance scan was the first event (~2247 hrs). The attacker mapped the network before exploiting any service. This is the standard cyber kill chain: Reconnaissance → Weaponization → Delivery.

**2. Which evidence proves credential theft.** Wireshark captures are the primary evidence. The FTP stream clearly shows USER and PASS commands in plaintext. The Telnet stream shows keystroke-by-keystroke login. The HTTP POST from DVWA shows form fields including username and password in the packet payload.

### 3. How to contain the incident in 10 minutes?

3.1 00:00 — Isolate compromised host 10.10.10.1 (remove from network).
3.2 00:02 — Block 203.0.113.99 at all perimeter ACLs and firewall rules.
3.3 00:04 — Kill malicious process PID 3421, preserve forensic image.
3.4 00:05 — Reset all credentials for FTP, Telnet, and DVWA immediately.
3.5 00:07 — Restore correct OSPF routing configuration on R1.
3.6 00:09 — Activate enhanced Suricata monitoring on all segments.

### 4. Long-Term Remediation Plan

4.1 Replace FTP with SFTP (SSH File Transfer Protocol) — encrypts both commands and d**ata.**
4.2 Replace Telnet with SSH — encrypted remote management.
4.3 Enforce HTTPS on DVWA — prevent credential sniffing from web logins.
4.4 Deploy OSPF MD5 or SHA authentication on all routers.

4.5 Implement network segmentation using VLANs — servers in dedicated DMZ.
4.6 Deploy EDR (Endpoint Detection and Response) on all hosts.
4.7 Configure egress ACLs to block outbound connections on non-standard ports.
4.8 Regular penetration testing and vulnerability assessments.

## 5. Which protocols should be replaced?

| Insecure Protocol | Replacement | Reason |
|---|---|---|
| FTP (Port 21) | SFTP (Port 22) | FTP sends credentials and data in plaintext |
| Telnet (Port 23) | SSH (Port 22) | Telnet sends every character unencrypted |
| HTTP (Port 80) | HTTPS (Port 443) | HTTP POST credentials visible in traffic |
| OSPF (no auth) | OSPF + MD5 Auth | Unauthenticated routing enables injection |

## 6. How to Improve Monitoring Posture

6.1 Deploy centralized SIEM (e.g., Wazuh, ELK Stack) to aggregate Suricata, system logs, and router logs.
6.2 Enable NetFlow/sFlow on all routers — detect anomalous traffic volumes.
6.3 Configure Suricata in IPS mode (inline) instead of passive IDS to actively block threats.
6.4 Create alerting thresholds: >50 connections/minute from a single host = automatic ACL drop.
6.5 Implement honeypots on legacy Telnet/FTP ports to detect future probing.